FRONT OIL SEAL REPLACEMENT [SKYACTIV-D 1.5]

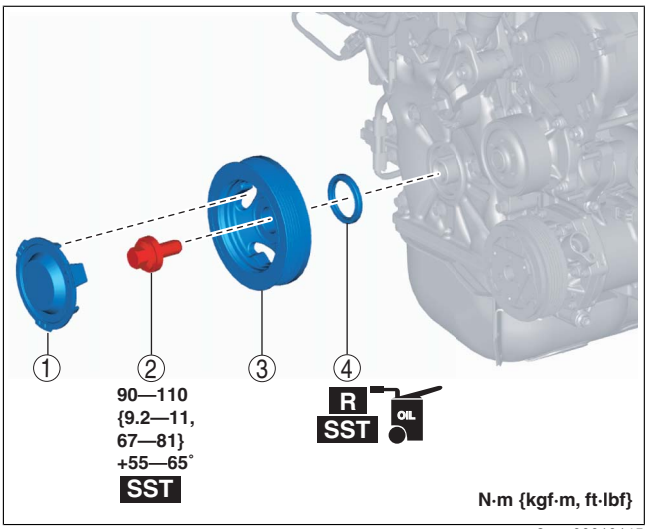
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Warning

- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before servicing.**

1. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
2. Remove the front splash shield. (RH) (See SPLASH SHIELD REMOVAL/INSTALLATION.)
3. Remove the drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Remove in the order indicated in the table.
5. Install in the reverse order of removal.

1	Crankshaft pulley cover (See Crankshaft Pulley Cover Installation Note.)
2	Crankshaft pulley lock bolt (See Crankshaft Pulley Lock Bolt Removal Note.) (See Crankshaft Pulley Lock Bolt Installation Note.)
3	Crankshaft pulley
4	Front oil seal (See Front Oil Seal Removal Note.) (See Front Oil Seal Installation Note.)



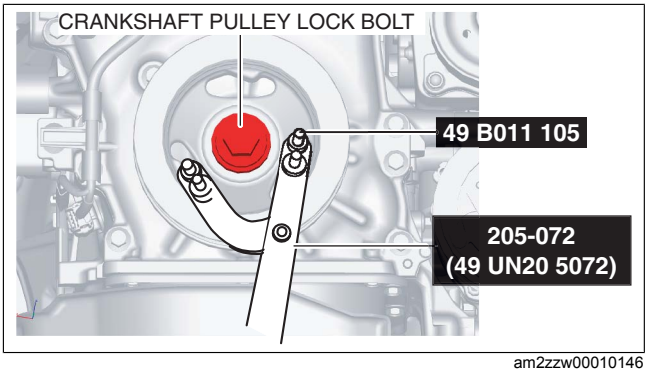
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Crankshaft Pulley Lock Bolt Removal Note

Caution

- **To prevent damage to the crankshaft pulley, protect the crankshaft pulley with a clean cloth so that the bar part (205-072) of the SST does not contact the crankshaft pulley.**
- **To prevent damage to the engine front cover, be careful not to allow the tab part (49 B011 105) of the SST to get caught between the crankshaft pulley and the engine front cover.**

1. Hold the crankshaft pulley using the **SSTs**.
2. Remove the crankshaft pulley lock bolt.

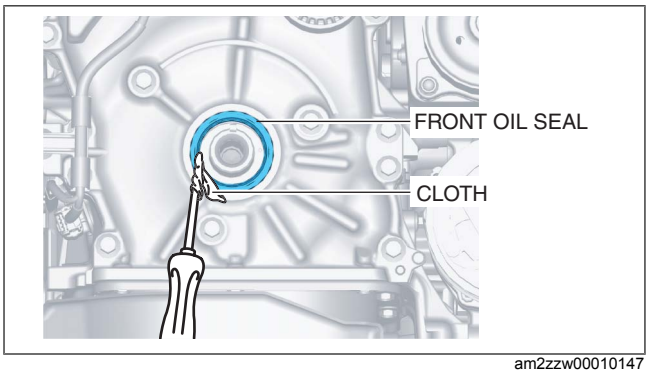


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Front Oil Seal Removal Note

1. Cut the oil seal lip using a utility knife.

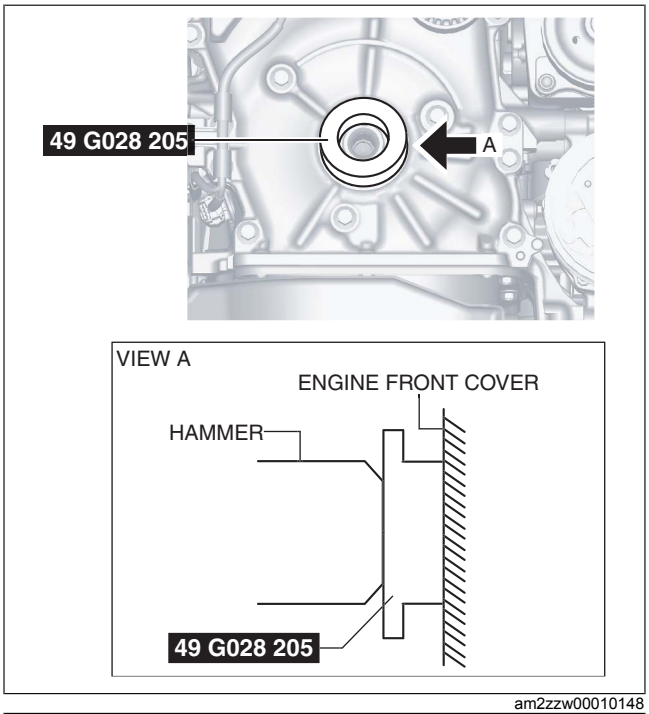
2. Remove the oil seal using a flathead screwdriver with the tip protected by a clean cloth to prevent damage to the oil seal sliding part of the crankshaft.



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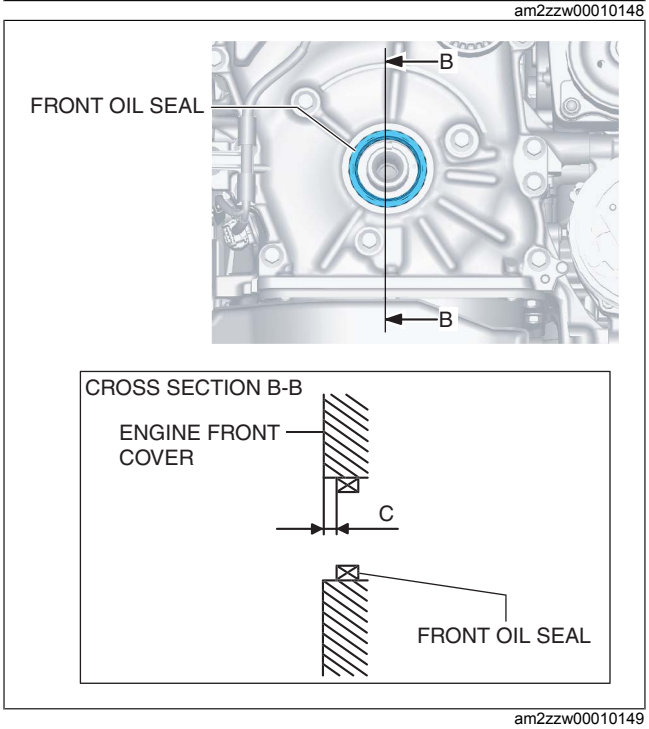
Front Oil Seal Installation Note

1. Apply clean engine oil to the inner surface of a new front oil seal.
2. Insert the front oil seal into the engine front cover by hand.
3. Tap the oil seal in evenly using the **SST** and a hammer.



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Front oil seal press on amount C
0—1.0 mm {0—0.039 in} from engine front cover



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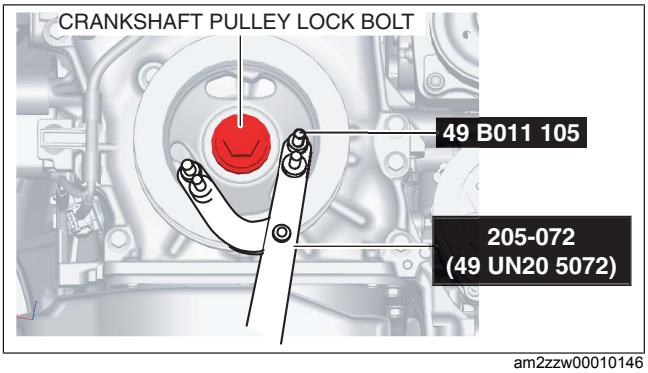
Crankshaft Pulley Lock Bolt Installation Note

Caution

- **To prevent damage to the crankshaft pulley, protect the crankshaft pulley with a clean cloth so that the bar part (205-072) of the SST does not contact the crankshaft pulley.**
- **To prevent damage to the engine front cover, be careful not to allow the tab part (49 B011 105) of the SST to get caught between the crankshaft pulley and the engine front cover.**

1. Hold the crankshaft pulley using the **SSTs**.
2. Tighten the crankshaft pulley lock bolt in the order shown in the following two steps.

Tightening torque
Step 1: 90—110 N·m {9.2—11 kgf·m, 67—81 ft·lbf}
Step 2: 55—65°



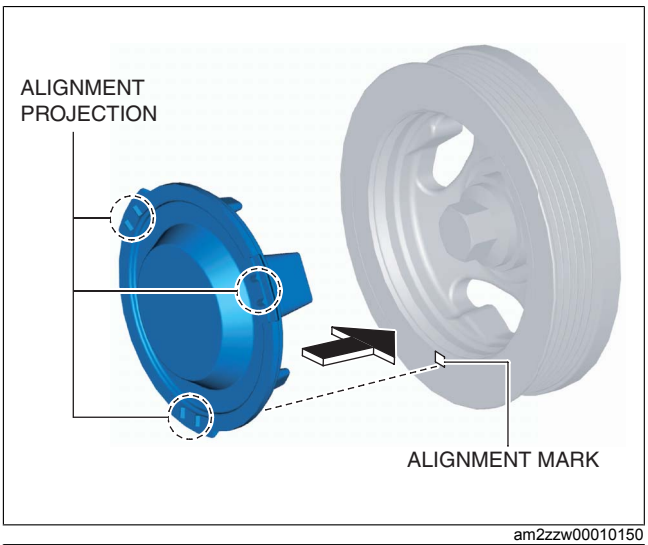
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Crankshaft Pulley Cover Installation Note

1. Align the crankshaft pulley alignment mark between the two projections on one of the three crankshaft pulley cover alignment projections.

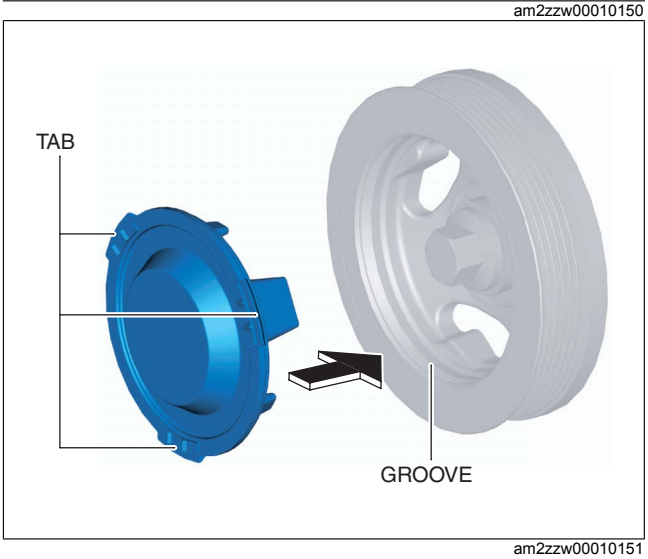
Note

- The alignment projections are in 3 locations, however any one of the three can be used for aligning.



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2. Press in the crankshaft pulley cover and attach the tabs of the pulley cover to the pulley groove.



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